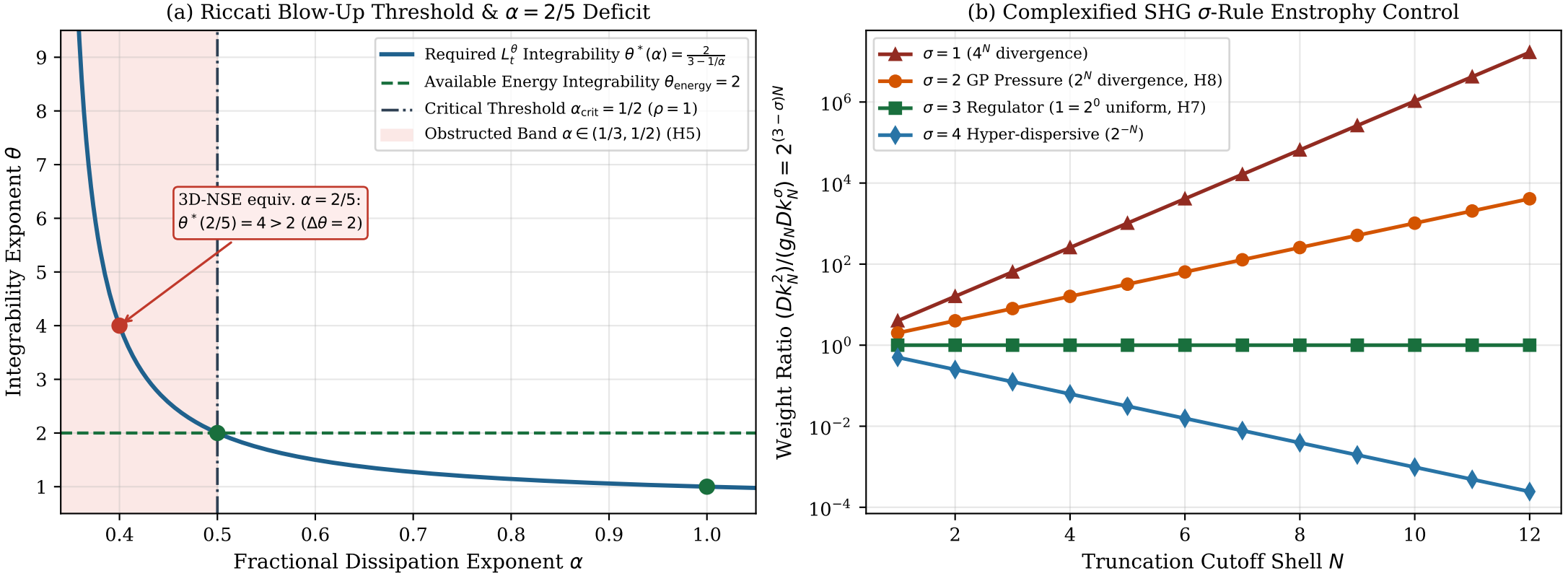
(a) Riccati Blow-Up Threshold & $\alpha = 2/5$ Deficit
Required L_t^θ Integrability $\theta^*(\alpha) = \frac{2}{3-1/\alpha}$
Available Energy Integrability $\theta_{\text{energy}} = 2$
Critical Threshold $\alpha_{\text{crit}} = 1/2$ ($\rho = 1$)
Obstructed Band $\alpha \in (1/3, 1/2)$ (H5)
3D-NSE equiv. $\alpha = 2/5$:
$\theta^*(2/5) = 4 > 2$ ($\Delta\theta = 2$)
Integrability Exponent θ
Fractional Dissipation Exponent α
(b) Complexified SHG σ-Rule Enstrophy Control
$\sigma = 1$ (4^N divergence)
$\sigma = 2$ GP Pressure (2^N divergence, H8)
$\sigma = 3$ Regulator ($1 = 2^0$ uniform, H7)
$\sigma = 4$ Hyper-dispersive (2^{-N})
Weight Ratio $(Dk_N^2)/(g_N D k_N^\sigma) = 2^{(3-\sigma)N}$
Truncation Cutoff Shell N